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Active

Build-your-own Weather App

Frontend Engineering Take-Home Assessment

Build a thoughtful single-page React prototype for a weather app that helps an outdoor meetup organizer compare this week's forecast to next week's, and communicate what to expect.

tl;dr

Build a React single-page prototype that lets an organizer:

- set an arbitrary event **location**,
- choose from **preset days** and **time ranges** (assume the event repeats weekly),
- view weather data (from the **Visual Crossing API**) with simple, human-friendly messages, and
- **compare this week's weather to next week's** (scroll when needed).

You have **72 hours** from receipt to submit a public URL to your implementation.

1 Overview

The objective is to showcase your ability to implement a thoughtful single-page prototype that is quick to iterate on. We recommend spending **4–6 hours** on the exercise.

2 Prompt

Given the following concept of a weather app for outdoor group events, how might you build a prototype that would test the viability of the design?

Feel free to take liberties where you feel it should be improved. The goal is to simulate an experience that can be experimented on quickly.

3 Constraints

3.1 User need

"When hosting a repeating Friday afternoon meetup in the park, I want to keep an eye on the weather forecast to let everyone know what to expect ahead of time so we can bring the right food, games, and gear. We'll often cancel this week if it's too wet or windy, and if next week's weather looks more favorable."

3.2 Data to show

Use any weather data from the Visual Crossing API. You will need to sign up to obtain an API key, but it is worth familiarizing yourself with before you begin coding.

Present generic weather message descriptions (examples):

- "nice day" if temperature falls between 60–75°F
- "chance of rain" if humidity falls between 25–75%

You may use any charting library (e.g., Chart.js or D3.js).

3.3 Interaction

Allow a meetup organizer to:

- set the event's arbitrary location,
- choose from preset days and time ranges (assuming the event repeats weekly), and
- scroll when needed.

3.4 Technology

The project should be built using React.

For ease of deployment, CodeSandbox is recommended, but any combination of source code + public link is acceptable.

Other technology choices (frameworks, styling tools, TypeScript vs. JavaScript, state management, data fetching, etc.) are entirely up to you. Use of LLM copilot tools is allowed (encouraged, even).

3.5 Look and feel

Use any design system, asset library, or component framework you feel comfortable with. You may also write your own styling.

(Some popular asset libraries: Material and Noun Project.)

3.6 Layout

See the reference sketch in Figure 1, which allows an organizer to compare this week's weather to the following week's weather.

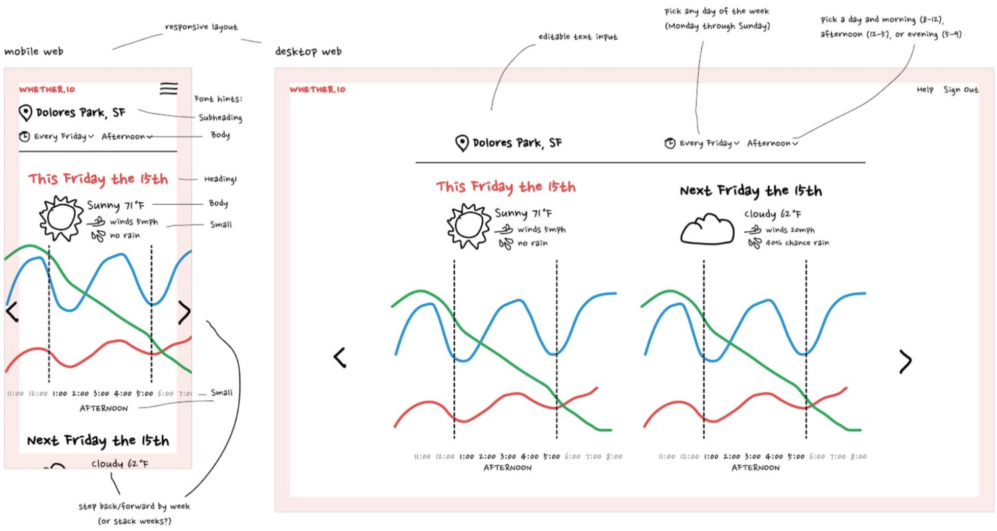

Figure 1: Reference sketch (mobile + desktop).

3.7 Time

You have **72 hours** from receiving this exercise to submit a URL to your CodeSandbox (or equivalent) where your project lives.

4 Evaluation

The expectation is that the product works and is appropriately styled. We are looking for a bit more than a bare-bones MVP. Your output will be evaluated against the following criteria:

1. **Design interpretation:** demonstrate an ability to understand the need of the user and the intent of the design. Make informed, calculated decisions when diverging from the design as needed.
2. **Software architecture:** build a product that reflects your perspective of good-enough software architecture with the right balance between tech debt and extensibility without boiling the ocean. Consider the flexibility of the application, such that your solution can evolve reasonably as business needs change.
3. **API integration:** demonstrate an ability to understand and integrate with public APIs involving potentially complex data models and error handling while maintaining a smooth user experience.

5 Deliverable

Your deliverable should be:

- a URL to your finished CodeSandbox, **or**
- a combination of source code + a publicly accessible app.

Regardless of platform, please use a free-tier product to host your implementation, and email the link to `submissions@novig.com` (CC the recruiter). Reach out if you have any questions.